

The Successful Integrated Efficiency: Maintaining Revenue Stream In Face of Demand *Shortfall* and Global Uncertainty by PT. Semen Indonesia Tbk.

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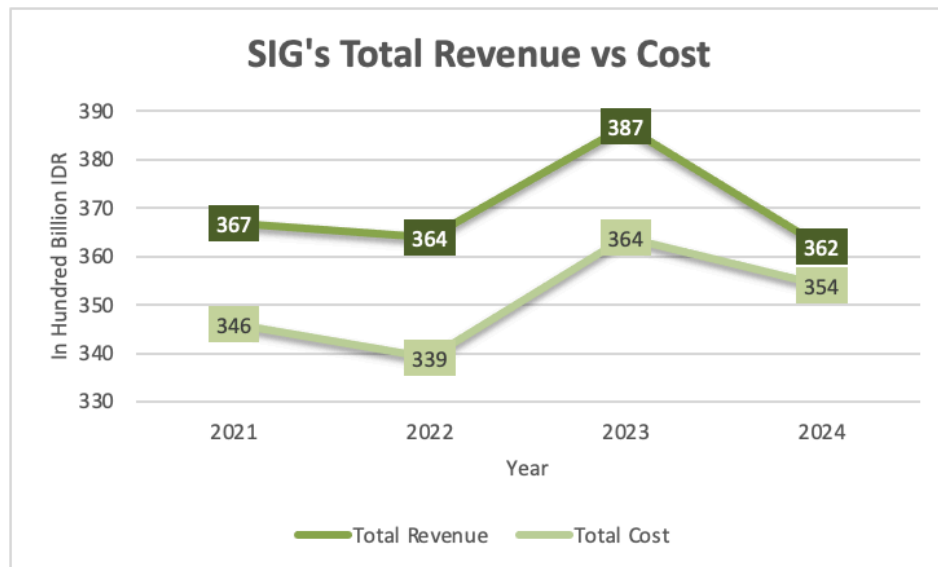
The Manufacturing Industry Hurdles



Source: Statista Market Insight; Writer estimates, proceed by author

The Indonesian manufacturing industry, which accounted for 18.98% of GDP in 2024, is facing significant challenges. Based on the IMF Article IV Indonesia 2025, the sector is constrained by three main issues: cost pressure, regional competitiveness, and uncertainty in the global economy. One segment within this industry, construction equipment manufacturing, reflects these pressures clearly. Since 2022, value-added growth in this sector has declined to below 8% and is projected to reach -1.4% in 2025, indicating stagnation and a slowdown in its long-term trajectory. Based on Arizton's report, competitiveness remains a major challenge for Indonesian companies, particularly due to the

influx of cheaper Chinese imports, while government policies have often favored the import of both used and new construction equipment for large infrastructure projects, largely bypassing local manufacturers. At the same time, structural constraints further complicate the landscape. Indonesia's geographical conditions contribute to persistently high logistics costs, creating a bottleneck for overall economic efficiency, as noted by Mantra Enterprise LLC. This is reflected in national data, where logistics costs remained high at 14.29% of GDP in 2022, according to the Coordinating Ministry for Economic Affairs.



Source: PT Semen Indonesia (Persero) Tbk (SIG) annual reports (2021-2024), proceed by author

In contrast to the challenges faced by the construction equipment manufacturing industry, PT Semen Indonesia (Persero) Tbk (SIG) presents an interesting case. While the sector has contracted since 2022, SIG maintained its revenue stream and financial stability through tighter cost control and efficiency measures. However, the situation is more complex than it appears. Based on its 2023 and 2024 annual reports, SIG's revenue declined from 38.65 trillion in 2023 to 36.19 trillion in 2024, indicating weakening demand. Theoretically, a demand shock should reduce revenue, yet this impact has been mitigated. The company implemented systematic efficiencies across production and distribution while maintaining a healthy balance sheet. This strategy is executed through an integrated system focusing on three areas: production, distribution, and financial (balance sheet) efficiency.

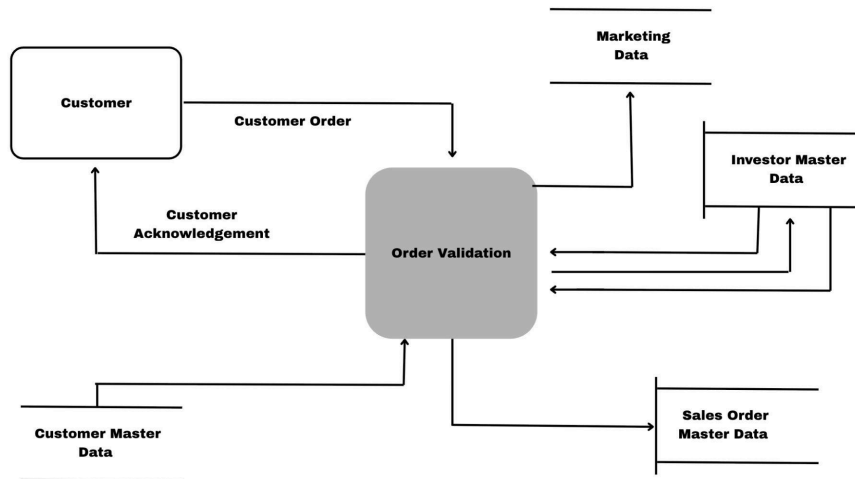
SIG departed from the conventional and fragmented formula for their case, and was designing a sequence which one operational improvement certainly would have a positive *domino effect* on the other sectors. The company believed that the increasing production efficiency would lower the cost burden at the distribution process. This phenomenon would strengthen the cash-flows for debt reduction and cementing a sustainable and healthy balance sheet for the next couple years. The following section elaborates comprehensively concerning how they did it.

The Integrated efficiency: The Structured and Cohesive Plan from Production and Distribution Sectors Improvements towards Balance Sheet Achievement

In implementing its strategy, SIG established a key principle: efficiency is “not a set of isolated solutions,” but rather something that must be executed across the entire business operation in a co-integrated manner. This approach has proven highly effective in responding to weakening domestic cement demand, where a “punch” to cash flow would otherwise amplify financial strain caused by inefficiencies. Adopting an end-to-end system perspective, the company recognizes the significant impact of energy consumption, particularly electricity, and has successfully deployed Plant Optimizer technology, which was recognized by the Ministry of Industry through the INDI 4.0 Award 2024 in the sustainable technology category. For example, in Tuban (East Java), SIG has implemented Advanced Process Control, Quality Control through QCX Robolab, and Waste Heat Recovery Power Generation (WHRPG), resulting in an 8% increase in production capacity, a 4.3% reduction in production costs, electricity savings of 4 kWh per ton, and a reduction in emissions by 122,358 tons per year.

On the other hand, energy diversification is used to tackle cost volatility. The impact of the combination policy between Refuse-Derived Fuel (RDF), which had been used for a decade, named the “Pilot Project,” and the new 6.5 MWp solar panel installations (increased from 0.12 MWp in 2023) is not only reducing emissions to 1,450,260 kg CO₂ but also supplying 60–70% of their laboratory energy needs. This situation indicates the capability not only to adapt and catch up with current sustainable technology trends but also to maintain energy security and lower dependence on non-renewable resources. Furthermore, they extend their cost efficiency through clinker factor substitution. Clinker is the most energy-intensive component and is the main driver of cost. The clinker ratio decreased from 69.07% to 67.07%, thus lowering the Cost of Goods Sold (COGS).

Through the combination of implemented sustainable technologies, including Plant Optimizer technology for energy consumption, Refuse-Derived Fuel (RDF) and solar panel installations for energy diversification, and clinker factor substitution, SIG has achieved satisfactory electricity cost and raw material efficiencies. Based on their annual reports in 2023 and 2024, they were able to reduce the Specific Electrical Energy Consumption (SEEC) from 85.9 kWh/ton to 84.38 kWh/ton of cement in 2024 and increase the use of renewable materials from 1.33 million tons to 1.47 million tons. However, SIG did not stop there; they recognized that even if production is optimized, it does not necessarily mean that a healthy cash flow circulation is maintained in the accounting books. After optimizing the production process, SIG then moved to secure their distribution process.



Source: Fidia, Macnico, P., et al (2022), proceed by author

In order to cope with Indonesia's logistics issues, SIG launched a new Digital Integrated Supply Chain management system using digital technology and Artificial Intelligence (AI). With the help of AI, the company is able to set distribution priorities based on real-time inventory stock status and generate automatic forecasts based on historical customer transaction data. The results are significant; for example, order fulfillment increased by 1.16% (equivalent to 118,000 tons of cement), and data consolidation efficiency improved, as tasks that previously took days can now be completed in just two hours, while also lowering the risk of stock-outs through a real-time system. The use of AI further strengthens the existing Enterprise Resource Planning (ERP) system. In this distribution framework, the ERP system plays a major role in managing the entire distribution and inventory process, including automatically informing employees when stock is running low (Delivery Stock Product or D1) and notifying them when buyers have paid deposits for orders, both using SAP and JD Edwards, which integrate finance, procurement, and transportation functions. In addition, real-time truck surveillance and identification are conducted using RFID and GPS to monitor expedition movements, while a shipment management system is used to ensure delivery schedules do not overlap and to connect sales orders to specific expeditions or trucks.

Besides technology in supporting an efficient distribution, SIG also developed a multimodal infrastructure. The company operates 30 packing plants, 9 specialized ports, and hundreds of buffer warehouses, which are highly connected and coordinated to avoid miscommunication. Moreover, to ensure uninterrupted distribution, SIG often uses sea routes, which have proven to be highly effective. At the same time, the company aligns its subsidiary entities to create logistics synergy, for example PT Sinergi Informatika Semen Indonesia (SISI). One of the operational initiatives is backhaul optimization in 2024, which prevents trucks from returning to the factory empty, instead requiring them to also carry raw materials such as fly ash, bottom ash, or alternative fuels. This phenomenon has helped reduce the logistics cost per ton of cement.

The result of efficient production and distribution led to 91% of plant optimizer utilisation exceeding its target which is only 90%. Not only in terms of utilisation but also in SGI's cash

balance. They are able to secure a strong cash balance of 4,6 trillion at the end of 2024, although at the same time paid debt in the amount of 4,9 trillion. However, the optimal condition of production and distribution does not mean that the company is doing well, SIG understands this deeply. Without good financial-balance sheet management and allocation, the success of the real sector (production and distribution efficiencies) becomes worthless. The relationship between the financial and real sector is unquestionably important. So, after completing the production and distribution process, SIG continues to improve their financial management.

In the 2024 annual report, SIG is determined to do deleveraging. Because of the increasing efficiency in the production and distribution process, the company is able to lower costs thus could save more in the cash account. The cash is used to pay high interest debt to manage risk in a high economic uncertainty, thus reducing interest debt from 15,8 trillion in 2023 to 10,87 trillion in 2024. This act is improving the debt to equity ratio from 37% in 2023 to just 25% in 2024. Furthermore, the company's decision to use deposit as the prerequisite of ordering cement, which means simultaneously decreasing the risk of bad debt expense thus also decreasing account receivable, nominally. This situation in the management accounting perspective, would significantly raise the velocity of the cash conversion cycle (CCC), which would cut the dependency on loanable funds for daily operations or consumption and investment. At the end, as the result of the debt management and cash flow achievement, the interest expense plummeted by 20,2% yoy and even still defending its market share at 48,2%. From here, we could see the current PT Semen Indonesia circumstance is prevalent and strong, which means that it could not only sustain but find opportunities to expand and develop.

In summary, the analysis of PT Semen Indonesia (SIG) shows a company capable of maintaining its revenue stream despite contracting cement demand and heightened geopolitical uncertainty. In facing such complex challenges, efficiency becomes the primary lever. Ideally, a firm can either reduce costs or increase revenue, but in this context SIG can mainly control costs, while revenue depends largely on external factors.

However, the issue cannot be addressed through isolated and fragmented solutions, but rather through a systematic approach. To implement such an approach, it is essential to understand the structure and sequence of actions to ensure that each step supports the next. Cost reduction must begin by identifying the root cause of costs. In this case, it lies in production, more specifically in energy and electricity. If energy prices or supply become volatile or inefficient, this creates a domino effect on production, distribution, and ultimately the balance sheet. In other words, if the core issue in energy is not resolved, efforts in distribution or financial allocation would have limited impact. By addressing the problem at its most fundamental level, subsequent improvements become effective. SIG demonstrates this understanding by investing in sustainable energy for the long term and improving efficiency through advanced technologies. Once the production process is secured, the company strengthens its distribution infrastructure and integrated systems, which further enhances performance. These combined efficiencies in production and distribution ultimately generate stronger cash flows, enabling debt repayment and improving financial resilience.

Conclusion and Key takeaway

Learning from the PT Semen Indonesia' s success gives us an essential lesson in solving a business case. Tackling a real world problem would not be as light as a fictional or competition based prompt. In doing so we could not use frameworks, concepts, and theories so strictly but combining them with our experience and imagination of how to implement the solution properly and feasibly, as we see from the SIG's strategy from using highly complicated ERP systems to use creativity for backhaul optimization.

From this case, we should develop “radical thinking skills” to find the root cause, SIG is able to figure out the importance of production optimization for distribution and financial sustainability. Moreover, we should sense the problem deeply so that we can formulate an effective, efficient, integrated, and cohesive solution in the most elegant manner.

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